2022 SSMO Speed Round

SMO Team

SR 1: Bobby is bored one day and flips a fair coin until it lands on tails. Bobby wins if the coin lands on heads a positive even number of times in the sequence of tosses. The probability that Bobby wins can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

SR 2: A bag is big enough to hold exactly 6 large pencils, 12 medium pencils, or 30 small pencils, with no space left over. Given that there is 1 large pencil and 3 medium pencils currently in the bag, what is the maximum number of small pencils that may be added to the bag?

SR 3: Let $ABCD$ be a parallelogram. Let E be a point on CD such that $\frac{CE}{DE} = \frac{2}{3}$. Suppose that BE and AC intersect at F and the area of triangle AEF is 15. Find the area of $ABCD$.

SR 4: Consider a quadrilateral $ABCD$ with area 120 and satisfying $AB + CD = AD + BC = 24$. There exists a point P in 3D space such that the distances from P to AB, BC, CD, and DA are all equal to 13. Find the volume of $PABCD$.

SR 5: Let $ABCD$ be a square such that E is on AD and F is on CD. If $AE = DF$ and $\frac{[BEF]}{[ABCD]} = \frac{7}{18}$, then the value of $\frac{EF^2}{BC^2}$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

SR 6: At the beginning of day 1, there is a single bacterium in a petri dish. During each day, each bacterium in the petri dish divides into $a > 1$ new bacteria, and $b \geq 1$ bacteria are added to the petri dish. For example, at the end of day 1, there are $a + b$ bacteria in the petri dish. If, at the end of day 4, the number of bacteria in the petri dish is a multiple of 48, find the minimum possible value of $a + b$.

SR 7: Let $A_1 = (1, 0)$. Define A_{n+1} as the image of A_n under a rotation of either 45°, 90°, or 135° clockwise about the origin, with each choice having a $\frac{1}{3}$ chance of being selected. Find the expected value of the smallest positive integer $n > 1$ such that A_n coincides with A_1.

SR 8: How many positive integers cannot be written as $7a + 19b + 28c$, where a, b, and c are positive integers (not necessarily distinct)?

SR 9: Consider a triangle ABC such that $AB = 13$, $BC = 14$, $CA = 15$ and a square $WXYZ$ such that Y and Z lie on $\overleftrightarrow{BC}$, W lies on $\overleftrightarrow{AB}$, and X lies on $\overleftrightarrow{CA}$. Suppose further that W, X, Y, and Z are distinct from A, B, and C. Let O be the center of $WXYZ$. If AO intersects BC at P, then the sum of all values of $\frac{BP}{CP}$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

SR 10: Let $S = \{2, 7, 15, 26, \ldots\}$ be the set of all numbers for which the ith element in S is the sum of the ith triangular number and the ith positive perfect square. Let T be the set which contains all unique remainders when the elements in S are divided by 2022. Find the number of elements in T.